Vacuum 162 (2019) 20–27

Contents lists available at ScienceDirect

Vacuum

journal homepage: www.elsevier.com/locate/vacuum

Oxidation response of a vacuum arc melted NbZrTiCrAl refractory high entropy alloy at 800–1200 °C

Ping Zhang[a,b,*], Yuantian Li[a], Zheng Chen[a,**], Jinyong Zhang[a], Baolong Shen[a,b]

[a] School of Material Science and Engineering, China University of Mining and Technology, Xuzhou, Jiangsu, 221116, PR China
[b] The Jiangsu Province Engineering Laboratory of High Efficient Energy Storage Technology and Equipment, PR China

ARTICLE INFO

Keywords:
High entropy alloy
Refractory metals
Vacuum arc melting
Oxidation kinetics
Oxidation products

ABSTRACT

An equimolar refractory high-entropy alloy NbZrTiCrAl was prepared by vacuum arc melting, the microstructure and oxidation response of the alloy at 800 °C, 1000 °C and 1200 °C were investigated. The alloy mainly consists of BCC single-phase solid solution and shows an inhomogeneous microstructure with dendritic and inter-dendritic regions. The oxidation of the alloy at 800 °C and 1000 °C follows parabolic rate law, and the protection performance can be attributed to the formation of dense and homogenous complex oxides scale composed of $CrNbO_4$, ZrO_2, TiO_2, Al_2O_3 and $ZrNb_2O_7$. The linear oxidation kinetics of the alloy at 1200 °C is observed, the scale composed of similar oxide products but with layered structure is probably the reason of the poor resistance to long-term oxidation at higher temperatures. The formation of the oxide scale and the oxidation-resistant mechanism of the alloy at different temperatures have been discussed.

1. Introduction

The refractory high-entropy alloys (RHEAs) proposed by Senkov et al. at 2010, usually consisting of four or more refractory metal elements such as Nb, Mo, W, Ta, V, Hf, Ti, Zr and Cr in an equimolar ratio or close to equimolar ratio, have attracted wide attention as the potential high temperature structural materials [1–3]. Up to now, many studies emphasized on developing high-temperature performance of RHEAs have been reported, and most of the designed RHEAs possess outstanding strength and hardness, high thermal stability and high softening resistance at elevated temperatures [4–17]. NbMoTaW and VNbMoTaW with high hardness of 4.46 GPa and 5.42 GPa respectively show yield strength above 400 MPa at temperatures up to 1600 °C, and high structure stability is also observed in both alloys even at 1400 °C [1,5]. The compressive yield strength of $HfNbTiVSi_{0.5}$ alloy is as high as 1399 MPa, with a fracture strain of 10.9% at room temperature, and the yield strength is still preserved at 875 MPa and 240 MPa at 800 °C and 1000 °C respectively [9]. HfMoNbTaTiZr alloy possess the compression yield strength of 556 MPa at 1200 °C and with a fracture stain of 12% at room temperature [12]. By contrast, the study focus on high temperature oxidation behavior of the RHEAs is limited, although these alloys are designed for high temperature applications.

It is common knowledge that the oxidation resistance of the refractory metals at high temperatures is extremely poor due to their high chemical activity with respect to oxygen [3]. The poor oxidation resistance of the RHEAs is also a serious potential barrier to the long-term application of the alloys at elevated temperatures. $NbCrMo_{0.5}Ta_{0.5}TiZr$ alloy follows parabolic oxidation kinetics at 1000 °C for 100 h, forming a complex oxides scale but spalling from the alloy after oxidation [18]. The linear oxidation kinetics of the $NbCrMoTiAl_{0.5}$, $NbCrMoVAl_{0.5}$, $NbCrMoTiVAl_{0.5}$ and $NbCrMoTiVAl_{0.5}Si_{0.3}$ alloys at 1300 °C is observed, and it is found that the oxidation resistance of the alloys can be improved with Ti and Si addition, but reduced with V addition [19]. Although a parabolic oxidation kinetics is found in the AlCrMoTiW alloy at 1000 °C for 40 h, an inhomogeneous and porous oxide scale forms and exhibits a tendency to spalling [20]. The oxidation kinetics of the AlCrMoNbTi alloy follows the linear rate law at 900–1100 °C, while its oxidation resistance can be improved by addition of 1 at.% Si to the alloy [21]. The oxidation of the NbTiZrV alloy is rapid and exhibits linear behavior at 1000 °C, the improvement to the oxidation resistance of NbTiZrCr alloy is observed and attributed to formation of the complex oxides composed of $CrNbO_4$ and ZrO_2 [22]. The oxidation behavior of the NbTiZrAl alloy at 600–1000 °C for 50 h has been revealed, the oxide scale composed of simple oxides ZrO_2, Al_2O_3 and NbO forms at 600 °C, while composed of complex oxides $AlNbO_4$ and Ti_2ZrO_6 forms and reduces the oxidation rate at 1000 °C [23].

Refractory metals W, Mo, Ta and Nb possess high melting points and are considered as prime candidates for the RHEAs system. Compared

* Corresponding author. School of Material Science and Engineering, China University of Mining and Technology, Xuzhou, Jiangsu, 221116, PR China.
** Corresponding author.
E-mail addresses: zhangping0095@cumt.edu.cn (P. Zhang), chenzheng1218@163.com (Z. Chen).

https://doi.org/10.1016/j.vacuum.2019.01.026
Received 17 December 2018; Received in revised form 8 January 2019; Accepted 17 January 2019
Available online 18 January 2019
0042-207X/ © 2019 Elsevier Ltd. All rights reserved.

with W, Mo and Ta, Nb is more attractive due to its lower density and the formation of more stable oxides at high temperatures [24]. To further reduce the density of the alloy system, Zr (ρ_{Zr} = 6.51 g/cm^3) and Ti (ρ_{Ti} = 4.51 g/cm^3) can be added [4,24,25]. In addition, alloying with Ti effectively improves both the strength and ductility of the RHEAs at room temperature [26]. The introduction of Cr and Al are expected to form protective oxides on the alloy surface to ensure its oxidation resistance [4,19,22]. It is noted that the addition of Al could stabilize the bcc crystal structure and help to reduce the density of the RHEAs [27,28]. Vacuum arc melting is the most commonly used preparation method for high-entropy alloys [29,30]. Therefore, in this work, an equimolar NbZrTiCrAl refractory high entropy alloy is designed and prepared by vacuum arc melting, the microstructure and oxidation response of the alloy at 800 °C, 1000 °C and 1200 °C are investigated.

2. Experimental procedures

The equimolar NbZrTiCrAl refractory high entropy alloy ingot was prepared by vacuum arc melting in a pure argon atmosphere. The purity of the metal raw materials used in the preparation was greater than 99.9 wt%. During melting, the metal materials of each element were placed in the water-cooled copper crucible in descending order of melting point (Al was placed the bottom of the copper crucible, Nb was placed on the top). In addition, in order to make the composition of alloy ingot melted more homogeneous, each alloy ingot was re-melted by inversion 8 times, and the melting time was maintained at about 2 min each time.

In order to investigate the oxidation response of the NbZrTiCrAl refractory high entropy alloy at high temperatures, the oxidation test was carried out at 800 °C, 1000°C and 1200 °C respectively in air. Before the oxidation test, the alloy ingot was cut into 10 mm × 10 mm × 5 mm cubes by electro-discharge machining. All the surfaces of the specimens were ground using SiC papers up to #1200, cleaned in an ultrasonic acetone bath and then dried. The alloy specimen was placed in an alumina crucible so that the total mass change including spalled oxides could also be obtained. During the oxidation process, the mass changes of the alloy specimens (total mass of the specimen with possible spalled oxides contained in the crucible) with holding times of 5, 10, 25, 30 and 50 h respectively were weighed in an electronic balance with an accuracy of 0.0001 g. Three measurements for each specimen were conducted and the average value with standard deviation was given.

The phase constituents of the NbZrTiCrAl refractory high entropy alloy before and after oxidation were identified using X-ray diffraction (XRD, Bruker D8 Advance) with Cu Kα radiation. Working voltage and operating current of the test conditions were 40 KV and 30 mA, respectively. The X-ray diffraction wavelength was λ = 0.15418 nm, the scanning speed was 2 degree/min, the step length was 0.02 degree, and the scan angle range was 10°–90°. The microstructure and the chemical compositions of the alloy before and after oxidation were analyzed either using a scanning electron microscopy (SEM, SU3500) or an electron-probe microanalysis (EPMA, EPMA-8050G) with wavelength dispersive X-ray spectroscopy (WDX).

3. Results and discussion

3.1. Microstructure of the alloy

Fig. 1(a) shows the XRD pattern of the NbZrTiCrAl refractory high-entropy alloy in the as-cast state. The main phase of the alloy is BCC, while weak diffraction peaks of aluminide phases (such as Al_2Zr, Nb_2Al) are also found in the alloy. The microstructure of the alloy is presented in Fig. 1(b), and exhibits a typical dendrite and inter-dendrite structure. The chemical compositions of the corresponding phases presented in the microstructure are also listed in Fig. 1(b). In order to further understand the microstructure of the alloy, the EPMA element mapping analysis of the alloy is presented in Fig. 2. It is further verified that Nb and Ti are mainly distributed on dendrites, while Al, Cr, and Zr are mainly segregated in the inter-dendritic region.

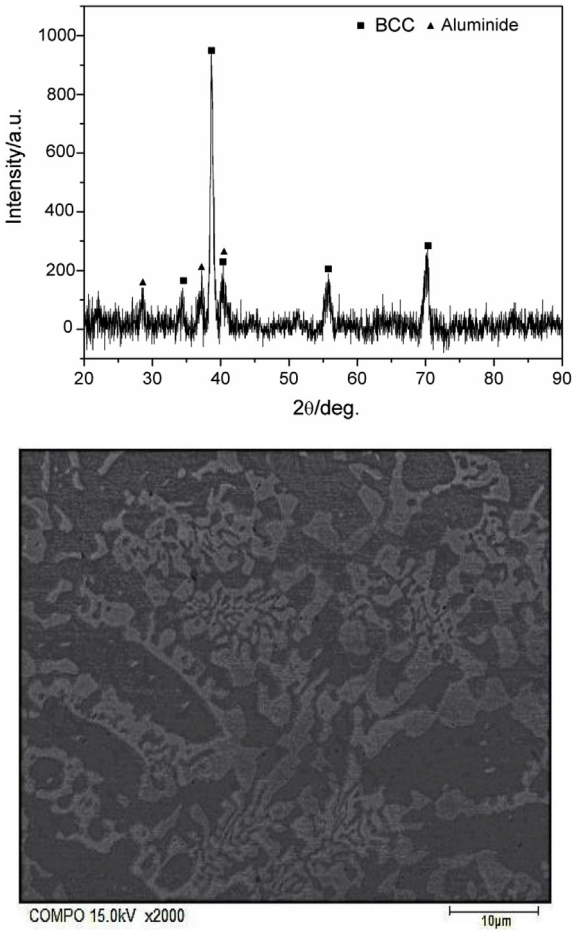


Fig. 1. XRD pattern and BSE microstructure of the NbZrTiCrAl alloy in as-cast state.

3.2. Oxidation kinetics of the alloy

Fig. 3 shows the oxidation kinetic curves and macroscopic photographs of the Nb ZrTiCrAl refractory high-entropy alloy being oxidized at 800 °C, 1000 °C and 1200 °C up to 50 h respectively. A slow mass gain of the alloy occurs during the whole oxidation process at 800 °C, and the oxide scale formed on the surface of the alloy after 50 h oxidation is still dense and complete. For the oxidation at 1000 °C, the mass gain of the alloy increases slowly in the initial 25 h but increases a bit fast after that. When being oxidized at an elevated temperature of 1200 °C, a sharply mass gain of the alloy is observed during the whole oxidation process. The macroscopic photographs of the alloy after oxidation at 1000 °C and 1200 °C also reveal that relative tightly adherent oxide scales formed on the surfaces of the alloy in the initial 25 h of oxidation, but cracking or spalling of the oxide scales can be observed after 25 h of oxidation during the rapid cooling process from the test temperature to room temperature in air. The mass gains behavior of the alloy during oxidation at 800, 1000 and 1200 °C for 50 h is illustrated by Power law ($\triangle m = kt^n$, where $\triangle m$ is mass gain per unit surface area (mg cm^{-2}), t is the holding time (h), k is the oxidation rate constant (mg cm^{-2} h^{-1}), and n is the time exponent), and the estimated k and n values are listed in Table 1. For the alloy being oxidized at 800 °C and 1000 °C for 50 h, the fitted n values are 0.57 and 0.64 respectively that are closer to 0.5, indicating the curves obtained at 800 °C and 1000 °C nearly obey the

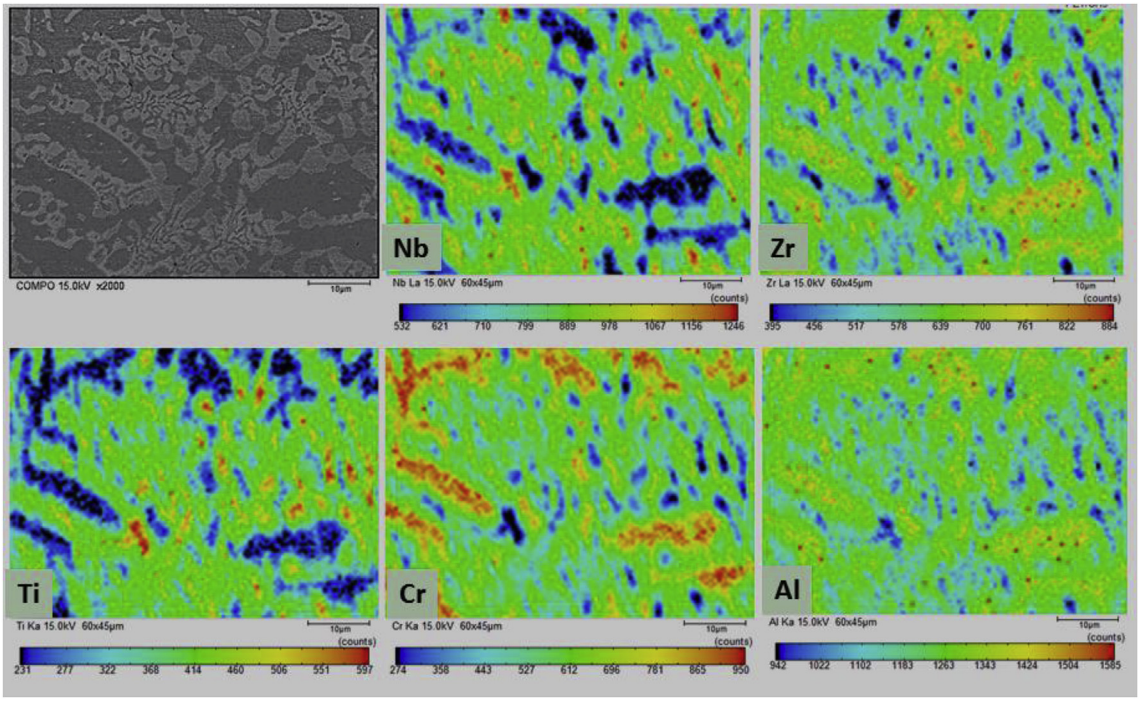


Fig. 2. EPMA element mapping of the NbZrTiCrAl alloy in as-cast state.

parabolic oxidation kinetics. However, for the alloy at 1200 °C for 50 h, the fitted n value is close to 1, revealing a linear oxidation kinetics. Obviously, it can be concluded that the alloy exhibits varying degrees of oxidation resistance at 800 °C and 1000 °C, but has poor resistance to long-term oxidation at 1200 °C.

3.3. Microstructure of the oxidized layer of the alloy

Fig. 4 shows the XRD patterns conducted on the surfaces of the NbZrTiCrAl refractory high-entropy alloy after oxidation at 800 °C, 1000 °C and 1200 °C for 5 h, 25 h and 50 h respectively. It can be seen that similar oxidation products formed on the surfaces of the alloy being oxidized at different temperatures with different holding times. $CrNbO_4$ is the predominant oxide in all the oxidation products of the alloy, and some amount of ZrO_2, TiO_2 and $ZrNb_2O_7$ are also present. In addition, the Al_2O_3 with relative weak peaks is also detected in the alloy after oxidation.

Fig. 5 shows cross-sectional BSE images of the NbZrTiCrAl refractory high-entropy alloy after oxidation at 800 °C for 5 h, 25 h and 50 h respectively. For the alloy after 5 h of oxidation, a 60 μm thick oxidized layer with dark contrast is observed in the cross-sectional BSE image of the alloy, as shown in Fig. 5(a). The oxidized layer consists of a 20 μm thick oxide layer and a 40 μm thick transition layer. WDX analysis reveals that the mean composition of the oxide layer is 67.5O-9.8Nb- 8.1Ti-5.9Zr-5.5Cr-3.2Al (at.%), and the mean composition of the transition layer is 29.9O-17.5Cr-15.8Zr-13.3Al-12.1Ti-11.4Nb (at.%). Combined with the surface XRD pattern in Fig. 4(a), the oxide layer can be determined to be the mixed oxides. The cross-sectional EPMA element mapping of the alloy after 5 h of oxidation is shown in Fig. 6. It can be seen that the Al-containing oxide (mainly Al_2O_3) is mainly distributed on the outer surface of the mixed oxides layer, indicating a selective oxidation of Al occurs, while a single Al_2O_3 protective layer is

Table 1
Fitted k and n values of the power law ($\triangle m = kt^n$) of the NbZrTiCrAl alloy during oxidation at different temperatures.

Oxidation temperature (°C)	k (mg cm^{-2} h^{-1})	n
800	1.36 ± 0.1	0.57 ± 0.02
1000	1.51 ± 0.7	0.64 ± 0.04
1200	7.8 ± 1.1	0.95 ± 0.12

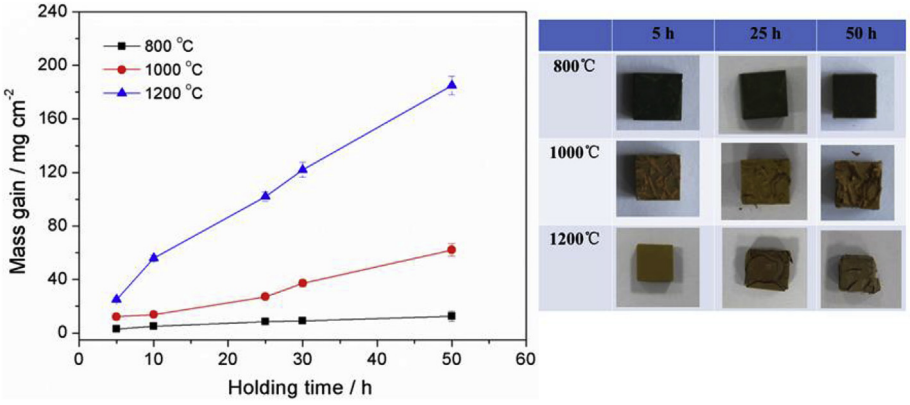


Fig. 3. Oxidation kinetic curves and macroscopic images of the NbZrTiCrAl alloy being oxidized at 800 °C, 1000 °C and 1200 °C respectively.

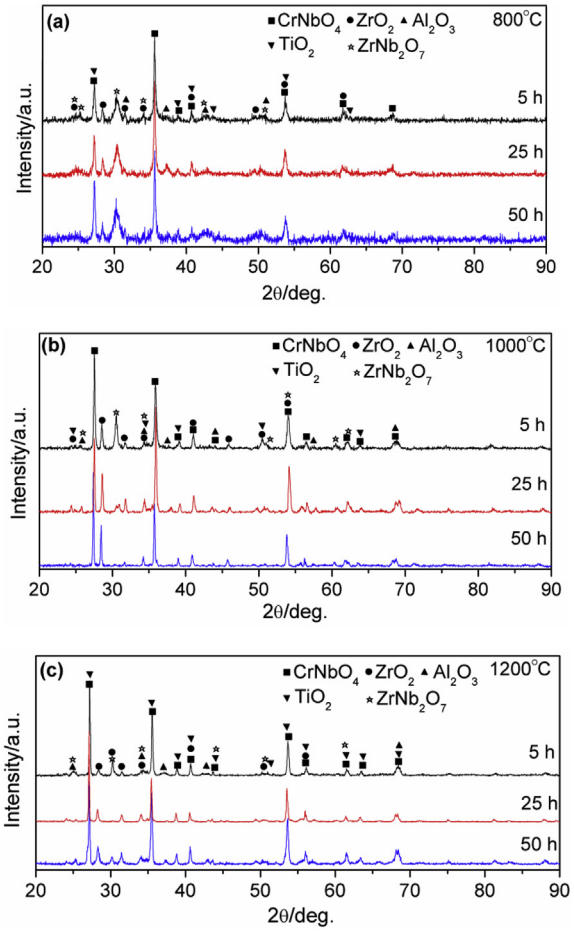


Fig. 4. XRD patterns of the NbZrTiCrAl alloy after oxidation at 800 °C (a), 1000 °C (b) and 1200 °C (c) for 5 h, 25 h, and 50 h respectively.

not formed, and the distribution of other oxides in the oxide layer is relatively uniform. In addition, although the oxidized layer formed on the surface of the alloy after 5 h of oxidation is a mixed oxides layer, the layer still exhibits good resistance to further diffusion of oxygen. For example, with increase in the oxidation time from 5 h to 25 h and 50 h, the thickness of the oxidized layer in the alloy increases slightly from 60 μm to 65 μm and 69 μm respectively, as shown in Fig. 5(b) and (c).

Fig. 7 shows cross-sectional BSE images of the NbZrTiCrAl refractory high-entropy alloy after oxidation at 1000 °C for 5 h, 25 h and 50 h respectively. After the alloy being oxidized at 1000 °C for 5 h, the oxidized layer is about 123 μm thickness and exhibits a distinct three-layer structure, as shown in Fig. 7(a). The outer oxide layer (an image of the magnified microstructure is inserted in Fig. 7(a)) with a thickness of 15 μm is uniform and dense, and WDX analysis indicates that the composition of the layer is 61.7O-10.3Zr-9.9Nb-7.4Cr-6.4Ti-4.4Al (at. %), which is also a mixed oxides layer. The thickness of the inner oxide layer is about 38 μm, and its composition is 41.3O-13.7Zr-12.9Cr-10.8Nb-8.7Al-7.5Ti (at.%) by WDX. The transition layer with an average oxygen content of 20.1 at.% is about 70 μm thick, and remains the dendrite structure of the original alloy. The EPMA element mapping of the alloy after oxidation at 1000 °C for 5 h is presented in Fig. 8. The cross-sectional analysis of the oxidized alloy also reveals that no protective oxide scale, neither Al_2O_3 nor Cr_2O_3, is formed on the alloy surface. While a discontinuous Al_2O_3 is visible at the outermost surface of the outer oxide layer, Cr-containing oxide is only found as a constituent of the outer oxide layer that represents a complex oxides. In addition, nitride is detected at the interface between the outer oxide layer and the inner oxide layer, as shown in Fig. 7(a). Further WDX analysis indicates that the nitride is mainly ZrN. As the oxidation time is further extended to 25 h and 50 h, as shown in Fig. 7(b) and (c), the oxidized layer of the alloy is thickened significantly, such as the thickness of the outer oxide layer is increased to 23 μm and 75 μm respectively, and the thickness of the inner oxide layer is increased to 105 μm and 120 μm respectively. The results of WDX analysis also reveal that the oxygen content in the oxidized layer decreases gradually from the outside to the inside, indicating that the thickening of the oxidized layer is mainly the result of the inward diffusion of oxygen.

Fig. 9 shows cross-sectional BSE images of the NbZrTiCrAl refractory high-entropy alloy after oxidation at 1200 °C for 5 h and 50 h respectively. The oxidized layer with a total thickness of about 315 μm is observed in the alloy after even 5 h of oxidation, as shown in Fig. 9(a). Compared with the relatively dense and homogeneously outer oxide layer formed on the surface of the alloy after oxidation at 800 °C

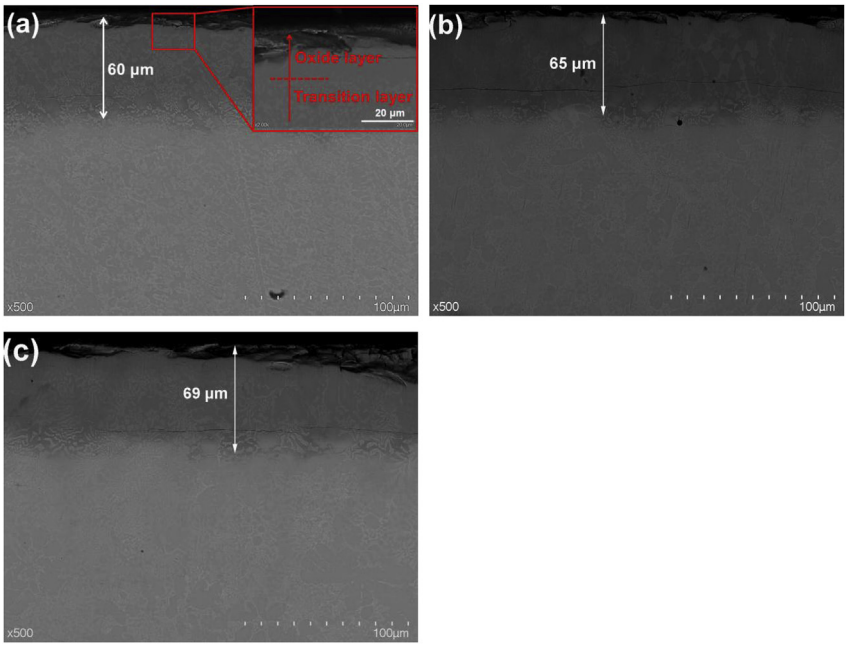


Fig. 5. Cross-sectional BSE images of the NbZrTiCrAl alloy after oxidation at 800 °C for 5 h (a), 25 h (b) and 50 h (c) respectively.

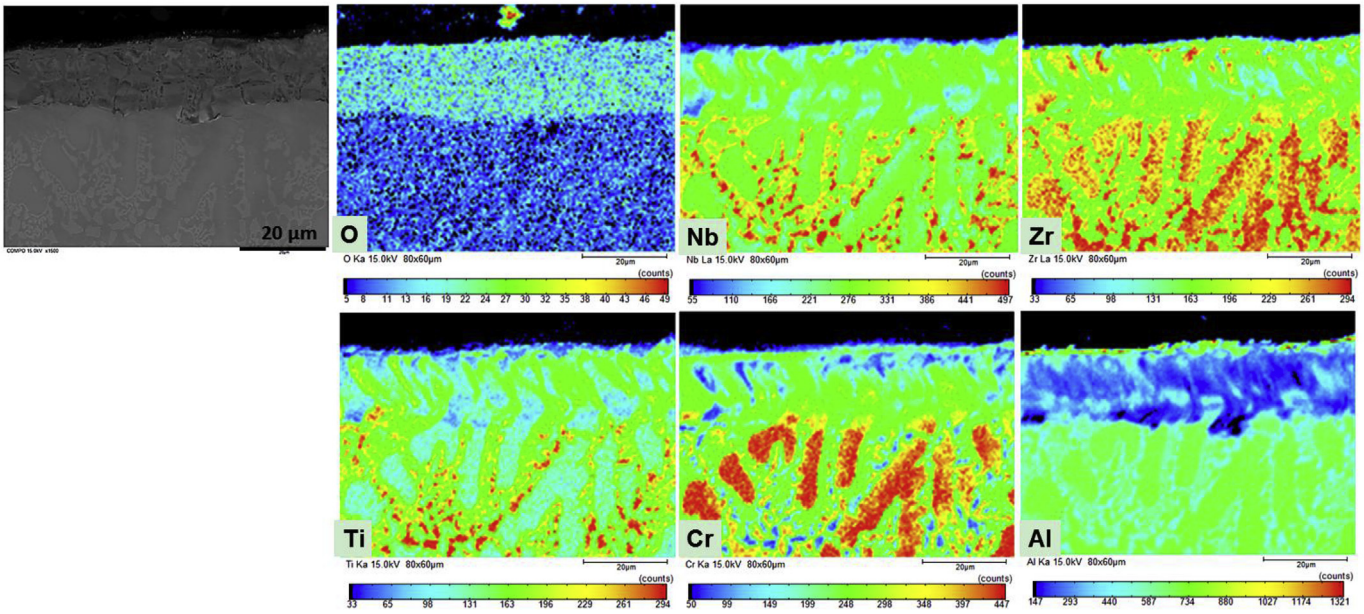


Fig. 6. Cross-sectional EPMA element mapping of the NbZrTiCrAl alloy after oxidation at 800 °C for 5 h.

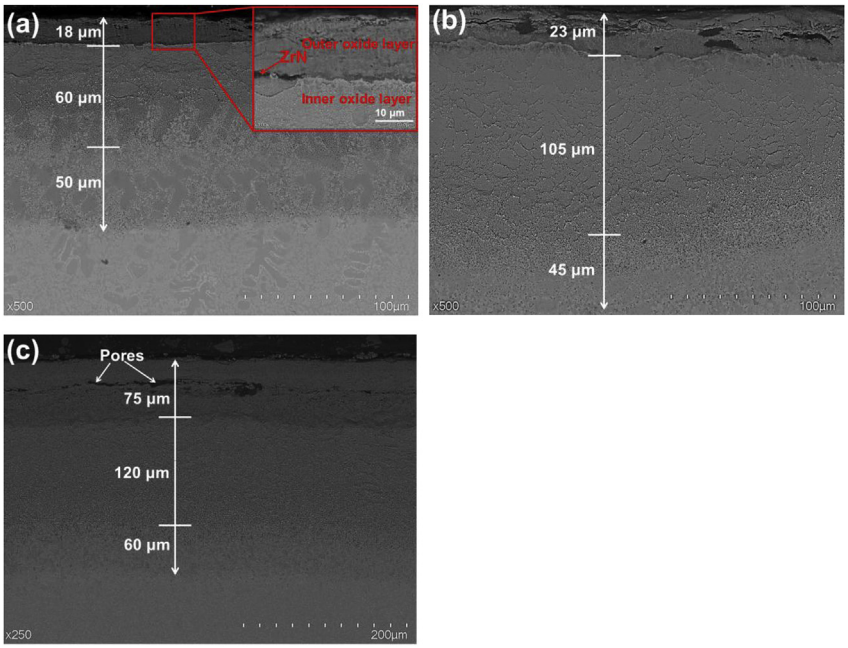


Fig. 7. Cross-sectional BSE images of the NbZrTiCrAl alloy after oxidation at 1000 °C for 5 h (a), 25 h (b) and 50 h (c) respectively.

and 1000 °C for 5 h, the outer oxide layer of the alloy after oxidation at 1200 °C for 5 h possess a distinct layered structure with textured porosity. Further EPMA element mapping analysis (in Fig. 10) also clearly reveals that the outer oxide layer is composed of various oxides layer by layer, such as Ti-oxide, Cr-Nb-oxide, Zr-oxide and Al-oxide from the outside to the inside. It is obvious that the outer oxide layer with the layered structure could not effectively prevent the inward diffusion of oxygen, and the weak bonding strength between the interfaces of the sub-layers leads to cracks easily. Furthermore, with the increase of oxidation time, the thick oxidized layer concentrates a large amount of tissue growth stress and thermal stress for the growth of oxides. This will cause the formed oxidized layer to be easily cracked in the subsequent oxidation process, and then a large number of oxygen diffusion channels generate which facilitates oxygen to quickly diffuse into the interior. As the oxidation time is prolonged to 50 h, a 1.8 mm thick oxidized layer formed in the alloy and is also partly shelled off during the cooling process from 1200 °C to room temperature, as indicated by the transversal cracks in the oxidized layer, as shown in Fig. 9(b).

3.4. Discussion

From Figs. 5, 7 and 9, it can be concluded that the NbZrTiCrAl alloy in this work exhibits mainly internal oxidation at 800–1200 °C. Similar oxidation mechanism has also been revealed in other RHEAs such as NbTiZrCr and NbTiZrV alloys [22]. It is elucidated that most of the elements containing in the RHEAs (e.g. Ti, Zr, Nb) possess high solubility for oxygen at elevated temperatures, combined with the lack of a protective external scale (such as Al_2O_3 or Cr_2O_3), which subsequently allows for the rapid ingress of oxygen [22]. Since all elements in the NbZrTiCrAl alloy possess approximately same concentration, the

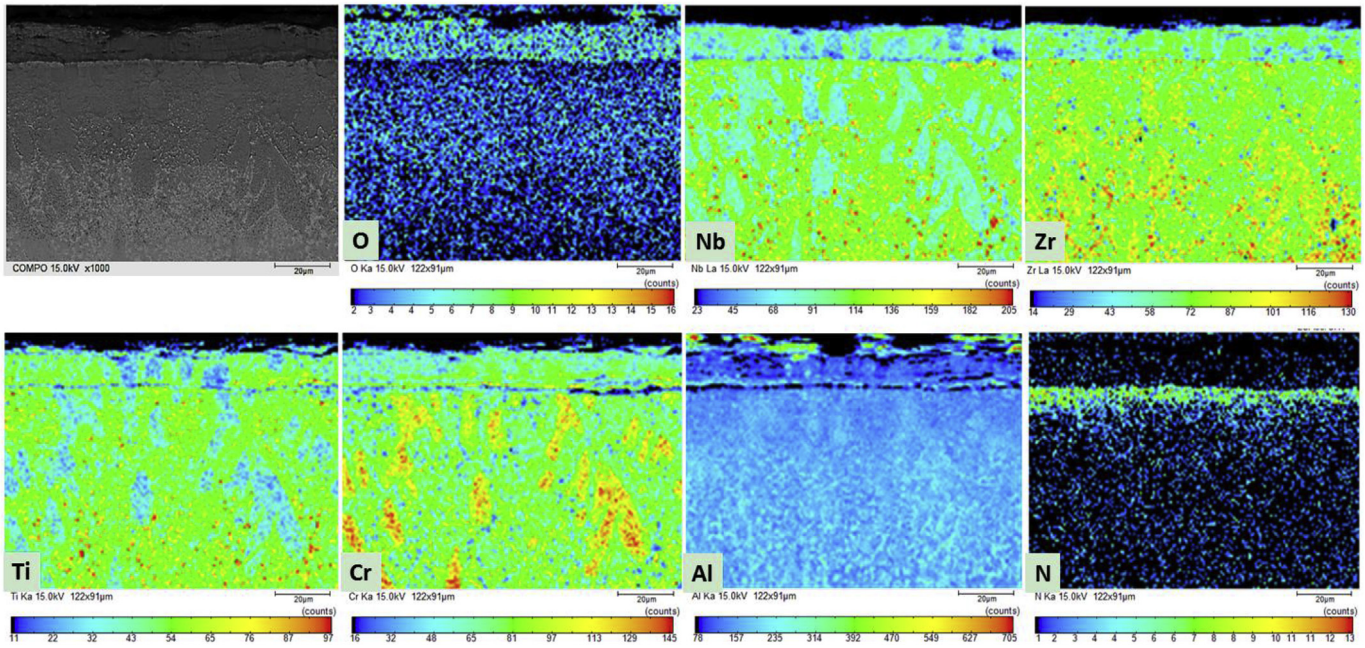


Fig. 8. Cross-sectional EPMA element mapping of the NbZrTiCrAl alloy after oxidation at 1000 °C for 5 h.

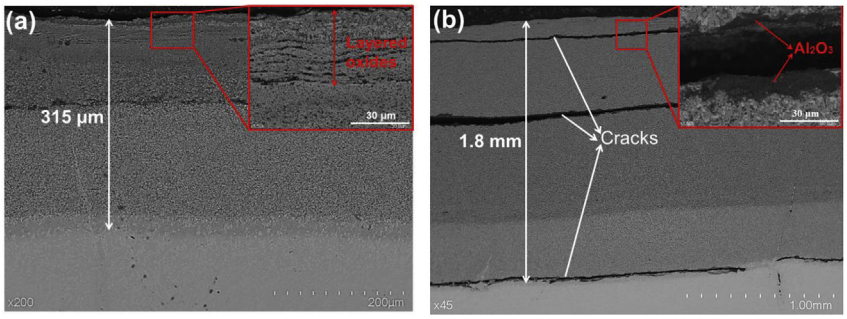


Fig. 9. Cross-sectional BSE images of the NbZrTiCrAl alloy after oxidation at 1200 °C for 5 h (a) and 50 h (b) respectively.

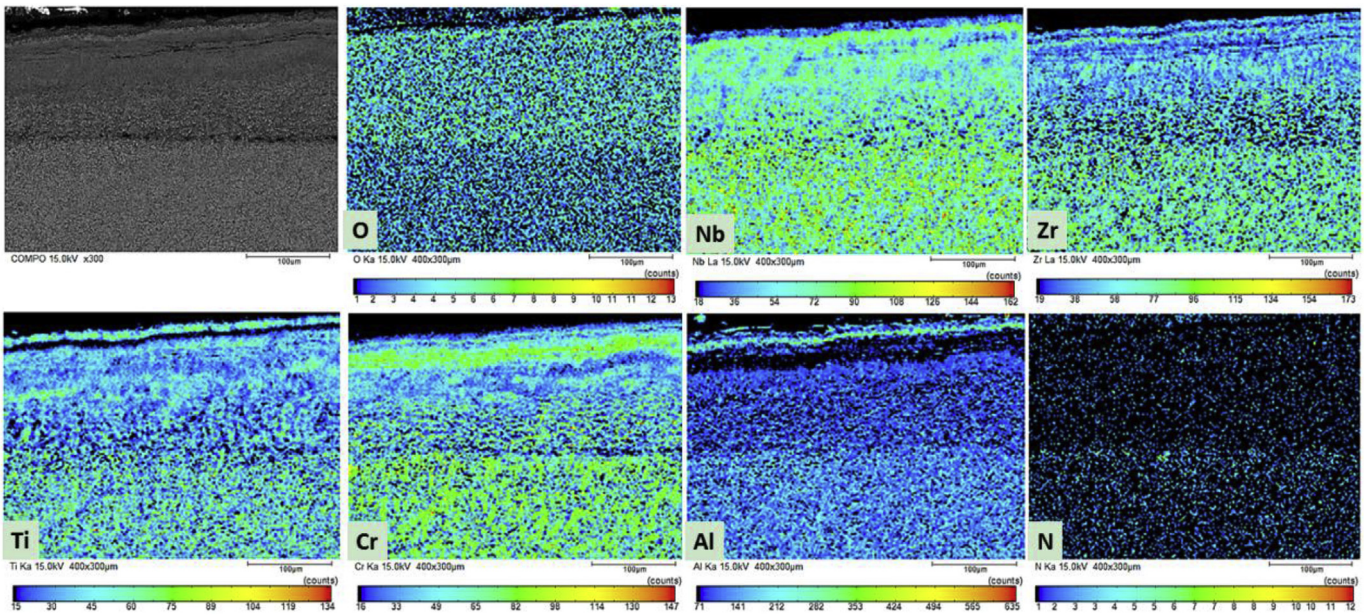


Fig. 10. Cross-sectional EPMA element mapping of the NbZrTiCrAl alloy after oxidation at 1200 °C for 5 h.

Table 2
Standard free energies of formation of oxides at 800 °C, 1000 °C and 1200 °C.

Temperature (°C)	ΔG^{θ}(KJ/mol, O_2)				
	ZrO_2	Al_2O_3	TiO_2	Nb_2O_5	Cr_2O_3
800	−896	−891	−750	−573	−569
1000	−859	−847	−714	−540	−535
1200	−822	−803	−679	−507	−502

Table 3
Standard free energies of formation of nitrides at 1000 °C and 1200 °C.

Temperature (°C)	ΔG^{θ}(KJ/mol, N_2)				
	ZrN	TiN	AlN	NbN	CrN
1000	−490	−435	−359	−229	−41
1200	−453	−398	−312	−193	−13

standard free energies for formation of the relevant oxides can be considered to assess their formation superiority and thermodynamic stability during oxidation. The standard Gibbs free energies of the most relevant oxides at 800–1200 °C are calculated using HSC Chemistry software and present in Table 2. It can be seen that the formation of favorable oxides follows the order as ZrO_2 > Al_2O_3 > TiO_2 > Nb_2O_5 > Cr_2O_3. From XRD analysis, it can be inferred that simple oxides (e.g. ZrO_2, TiO_2, Al_2O_3, Nb_2O_5 and Cr_2O_3) formed in the alloy at the initial stage of oxidation and then in most cases reacted to form more complex oxides, such as $CrNbO_4$ and $ZrNb_2O_7$. In addition, ZrN is also observed in the oxide scales of the alloy in Fig. 8, but it could not be identified by XRD patterns in the oxidation products of the alloy due to its little concentration. Due to the continuous thickening of the oxide layer, the oxygen partial pressure of the reaction between the matrix element and oxygen would be reduced. At this time, the relevant elements in the alloy may react with nitrogen to form nitrides in the scale. The standard Gibbs free energies for formation of relevant nitrides at 1000–1200 °C are also calculated using HSC Chemistry software, as shown in Table 3, and it is not surprising that ZrN as the most stable nitride in this alloy system is detected by EPMA element mapping in Fig. 8.

Although similar oxide products are found by XRD analysis in the complex oxides scales of the NbZrTiCrAl alloy after oxidation at 800–1200 °C, the structure of the complex oxide scales changes with increasing the temperature, resulting in different oxidation kinetics. The complex oxides scales mainly consisting of $CrNbO_4$ and ZrO_2 exhibit dense and homogenous structure in the alloy after oxidation at 800 °C and 1000 °C, while possess a porosity and layered structure in the alloy after oxidation at 1200 °C, as shown in Figs. 6, 8 and 10. The improvement to oxidation resistance of the NbZrTiCr alloy at 1000 °C has been reported by forming the complex oxides containing $CrNbO_4$ [22]. Similar sluggish oxidation kinetics are also observed in the NbZrTiCrAl alloy at 800–1000 °C, as shown in Fig. 3. As the oxidation temperature rises to 1200 °C, both diffusion rate of oxygen in the oxide scale and formation rate of the relevant oxides increase, which results in the preferred oxides in the alloy to be controlled not only by the thermodynamic stability but also by the growth kinetics. Thus, a complex oxides scale consisting same oxide products but with a layered structure is observed in the alloy after oxidation at 1200 °C, as shown in Fig. 10. It is clear that this complex oxides scale with weak bonding strength between the interfaces of the sub-layers could not effectively provide long term protection at high temperatures for the NbZrTiCrAl alloy.

4. Conclusion

In this paper, a novel refractory high entropy alloy NbZrTiCrAl was designed and prepared by vacuum arc melting. The alloy is composed of a BCC single-phase solid solution and exhibits a typical dendrite and inter-dendrite structure. Nb and Ti are mainly distributed on dendrites, while Al, Cr, and Zr are mainly segregated in the inter-dendritic region.

The alloy shows parabolic oxidation kinetics at 800 °C and 1000 °C, while linear oxidation kinetics at 1200 °C. The complex oxides scales mainly consisting of $CrNbO_4$ and ZrO_2 exhibit dense and homogenous structure in the alloy after oxidation at 800 °C and 1000 °C, while possess porosity and layered structure in the alloy after oxidation at 1200 °C. As the oxidation temperature rises, both diffusion rate of oxygen in the oxide scale and formation rate of the relevant oxides increase, which results in the preferred oxides in the alloy to be controlled not only by the thermodynamic stability but also by the growth kinetics. The alloy exhibits varying degrees of oxidation resistance at 800 °C and 1000 °C, but has poor resistance to long-term oxidation at 1200 °C.

Acknowledgements

This work is financially supported by the National Natural Science Foundation of China (Nos. 51601219, 51771226), the Natural Science Foundation of Jiangsu Province, China (No. BK20160261) and the Fundamental Research Funds for the Central Universities (No. 2015QNA08).

References

[1] O.N. Senkov, G.B. Wilks, D.B. Miracle, C.P. Chuang, P.K. Liaw, Refractory high-entropy alloys, Intermetallics 18 (2010) 1758–1765.
[2] D.B. Miracle, O.N. Senkov, A critical review of high entropy alloys and related concepts, Acta Mater. 122 (2017) 448–511.
[3] W.D. Li, P.K. Liaw, Y.F. Gao, Fracture resistance of high entropy alloys: a review, Intermetallics 99 (2018) 69–83.
[4] J. Chen, X.Y. Zhou, W.L. Wang, B. Liu, Y.K. Lv, W. Yang, D.P. Xu, Y. Liu, A review on fundamental of high entropy alloys with promising high-temperature properties, J. Alloy. Comp. 760 (2018) 15–30.
[5] O.N. Senkov, G.B. Wilks, J.M. Scott, D.B. Miracle, Mechanical properties of Nb25Mo25Ta25W25 and V20Nb20Mo20Ta20W20 refractory high entropy alloys, Intermetallics 19 (2011) 698–706.
[6] O.N. Senkov, C.F. Woodward, Microstructure and properties of a refractory NbCrMo0.5Ta0.5TiZr alloy, Mat. Sci. Eng. A 529 (2011) 311–320.
[7] O.N. Senkov, J.M. Scott, S.V. Senkova, D.B. Miracle, C.F. Woodward, Microstructure and room temperature properties of a high-entropy TaNbHfZrTi alloy, J. Alloy. Comp. 509 (2011) 6043–6048.
[8] O.N. Senkov, S.V. Senkova, C. Woodward, Effect of aluminum on the microstructure and properties of two refractory high-entropy alloys, Acta Mater. 68 (2014) 214–228.
[9] Y. Zhang, Y. Liu, Y.X. Li, X. Chen, H.W. Zhang, Microstructure and mechanical properties of a refractory $HfNbTiVSi_{0.5}$ high-entropy alloy composite, Mater. Lett. 174 (2016) 82–85.
[10] E. Fazakas, V. Zadorozhnyy, L.K. Varga, A. Inoue, D.V. Louzguine-Luzgin, F.Y. Tian, L. Vitos, Experimental and theoretical study of Ti20Zr20Hf20Nb20X20 (X= V or Cr) refractory high-entropy alloys, Int. J. Refract. Met. H. 47 (2014) 131–138.
[11] M.G. Poletti, G. Fiore, B.A. Szost, L. Battezzati, Search for high entropy alloys in the X-NbTaTiZr systems (X=Al, Cr, V, Sn), J. Alloy. Comp. 620 (2015) 283–288.
[12] C.C. Juan, M.H. Tsai, C.W. Tsai, C.M. Lin, W.R. Wang, C.C. Yang, C.M. Lin, S.K. Chen, S.J. Lin, J.W. Yeh, Enhanced mechanical properties of HfMoTaTiZr and HfMoNbTaTiZr refractory high-entropy alloys, Intermetallics 62 (2015) 76–83.
[13] J.P. Couzinie, L. Lilensten, Y. Champion, G. Dirras, L. Perriere, I. Guillot, On the room temperature deformation mechanisms of a TiZrHfNbTa refractory high- entropy alloy, Mat. Sci. Eng. A 645 (2015) 255–263.
[14] N.N. Guo, L. Wang, L.S. Luo, X.Z. Li, R.R. Chen, Y.Q. Su, J.J. Guo, H.Z. Fu, Hot deformation characteristics and dynamic recrystallization of the MoNbHfZrTi refractory high-entropy alloy, Mat. Sci. Eng. A 651 (2016) 698–707.
[15] H.W. Yao, J.W. Qiao, M.C. Gao, J.A. Hawk, S.G. Ma, H.F. Zhou, Y. Zhang, NbTaV-(Ti,W) refractory high-entropy alloys: experiments and modeling, Mat. Sci. Eng. A 674 (2016) 203–211.
[16] H.W. Yao, J.W. Qiao, J.A. Hawk, H.F. Zhou, M.W. Chen, M.C. Gao, Mechanical properties of refractory high-entropy alloys: experiments and modeling, J. Alloy. Comp. 696 (2017) 1139–1150.
[17] H.W. Yao, J.W. Qiao, M.C. Gao, J.A. Hawk, S.G. Ma, H.F. Zhou, MoNbTaV medium-entropy alloy, Entropy 18 (2016) 189.
[18] O.N. Senkov, S.V. Senkova, D.M. Dimiduk, C. Woodward, D.B. Miracle, Oxidation behavior of a refractory $NbCrMo_{0.5}Ta_{0.5}TiZr$ alloy, J. Mater. Sci. 47 (2012) 6522–6534.
[19] C.M. Liu, H.M. Wang, S.Q. Zhang, H.B. Tang, A.L. Zhang, Microstructure and oxidation behavior of new refractory high entropy alloys, J. Alloy. Comp. 583 (2014)

162–169.
[20] B. Gorr, M. Azim, H.-J. Christ, T. Mueller, D. Schliephake, M. Heilmaier, Phase equilibria, microstructure, and high temperature oxidation resistance of novel refractory high-entropy alloys, J. Alloy. Comp. 624 (2015) 270–278.
[21] B. Gorr, F. Mueller, H.-J. Christ, T. Mueller, H. Chen, A. Kauffmann, M. Heilmaier, High temperature oxidation behavior of an equimolar refractory metal-based alloy 20Nb20Mo20Cr20Ti20Al with and without Si addition, J. Alloy. Comp. 688 (2016) 468–477.
[22] T.M. Butler, K.J. Chaput, J.R. Dietrich, O.N. Senkov, High temperature oxidation behaviors of equimolar NbTiZrV and NbTiZrCr refractory complex concentrated alloys (RCCAs), J. Alloy. Comp. 729 (2017) 1004–1019.
[23] J. Jayaraj, P. Thirathipviwat, J. Han, A. Gebert, Microstructure, mechanical and thermal oxidation behavior of AlNbTiZr high entropy alloy, Intermetallics 100 (2018) 9–19.
[24] M.C. Gao, C.S. Carney, A.N. Dogan, P.D. Jablonksi, J.A. Hawk, D.E. Alman, Design of refractory high-entropy alloys, JOM 67 (2015) 2653–2669.
[25] O.N. Senkov, S.V. Senkova, C. Woodward, D.B. Miracle, Low-density, refractory multi-principal element alloys of the Cr-Nb-Ti-V-Zr system: microstructure and phase analysis, Acta Mater. 61 (2013) 1545–1557.
[26] Z.D. Han, N. Chen, S.F. Zhao, L.W. Fan, G.N. Yang, Y. Shao, K.F. Yao, Effect of Ti additions on mechanical properties of NbMoTaW and VNbMoTaW refractory high entropy alloys, Intermetallics 84 (2017) 153–157.
[27] C.M. Lin, C.C. Juan, C.H. Chang, C.W. Tsai, J.-W. Yeh, Effect of Al addition on mechanical properties and microstructure of refractory AlxHfNbTaTiZr alloys, J. Alloy. Comp. 624 (2015) 100–107.
[28] H. Chen, A. Kauffmann, B. Gorr, D. Schliephake, C. Seemüller, J.N. Wagner, H.-J. Christ, M. Heilmaier, Microstructure and mechanical properties at elevated temperatures of a new Al-containing refractory high-entropy alloy Nb-Mo-Cr-Ti- Al, J. Alloy. Comp. 661 (2016) 206–215.
[29] J. Zeng, C. Wu, H. Peng, Y. Liu, J. Wang, X. Su, Microstructure and microhardness of as-cast and 800°C annealed $Al_xCr_{0.2}Fe_{0.2}Ni_{0.6-x}$ and $Al_{0.2}Cr_{0.2}Fe_yNi_{0.6-y}$ alloys, Vacuum 152 (2018) 214–221.
[30] Y. Chen, S. Zhu, X. Wang, B. Yang, G. Han, L. Qiu, Microstructure evolution and strengthening mechanism of $Al_{0.4}CoCu_{0.6}NiSi_x$ (x = 0-0.2) high entropy alloys prepared by vacuum arc melting and copper injection fast solidification, Vacuum 150 (2018) 84–95.